RAGHAV UPADHYAY

AI / ML Engineer | LLMs · RAG · Deep Learning

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PROFESSIONAL SUMMARY

AI/ML Engineer building production-grade LLM systems — **RAG pipelines** with hybrid retrieval, cross-encoder reranking, hallucination detection, **LLM-as-a-judge evaluation**, and CI-gated quality thresholds. M.S. Data Science (May 2026, Univ. of Arizona). Hands-on with **Python, PyTorch, TensorFlow, Hugging Face Transformers, OpenAI API, LLaMA (Groq), ChromaDB**. Open to full-time AI Engineer roles in the US.

TECHNICAL SKILLS

LLMs & RAG: OpenAI API (GPT-4.1 family), LLaMA via Groq API, prompt engineering, vector embeddings, ChromaDB, sentence-transformers, hybrid search (vector + BM25), cross-encoder reranking, LLM-as-a-judge evaluation, hallucination detection, RAG pipelines, citation grounding

ML / Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, LSTM, attention mechanisms, deep ensembles, Monte Carlo Dropout, XAI / model explainability, uncertainty quantification

Languages: Python, SQL, R, JavaScript (basics), Bash

Data & Visualization: pandas, NumPy, NetworkX, Matplotlib, Seaborn, Jupyter, ipywidgets, Gradio

MLOps & Tools: Git, GitHub Actions (CI/CD), AWS, Linux, Hugging Face Spaces, Google Colab, REST APIs, Gradio, dotenv, virtualenv

PROFESSIONAL EXPERIENCE

Data Analyst Intern — Sudhir Mehrotra & Associates · Bareilly, India — *Jan 2025 – Aug 2025*

- Built Python ETL pipelines that automated financial workflows, cutting manual processing effort by **30%**.
- Developed a time-series cash-flow forecasting module that improved estimation accuracy by **15%** over baseline.
- Automated recurring Excel reporting with Python and VBA, eliminating multi-hour weekly manual tasks for the audit team.
- Designed structured data-reporting systems for audit and compliance teams, improving traceability across reviews.

PROJECTS

AskMyDocs — Production-Grade RAG System with CI-Gated Evaluation — *Apr 2026*

Python · LLaMA (Groq API) · ChromaDB · BM25 · Cross-Encoder · GitHub Actions · Gradio · Hugging Face Spaces

Live demo: huggingface.co/spaces/raghavupadhyay/askmydocs

- Shipped an end-to-end **RAG application** that answers natural-language questions over user-uploaded PDFs with grounded, citation-tagged responses, **deployed publicly on Hugging Face Spaces**.
- Engineered **hybrid retrieval** combining ChromaDB vector similarity (60%) and BM25 keyword scoring (40%), then re-scored top-10 candidates with a **cross-encoder reranker** (ms-marco-MiniLM-L-6-v2) to top-3.
- Implemented **hallucination detection** that flags any answer lacking inline citations, plus **versioned prompt templates** (default / strict / concise) selectable per query for behavior tuning.
- Built an automated **LLM-as-a-judge evaluation harness** (faithfulness, relevance, citation rate) wired into a **GitHub Actions CI pipeline** that fails the build when faithfulness < 0.70, relevance < 0.70, or citation rate < 0.80.
- Architected reusable, swappable modules: loader, chunker, embedder, vectorstore, reranker, llm, prompts, and pipeline.

LLM-Generated Social Networks — Cross-Cultural Study (Capstone) — *May 2026*

Python · OpenAI API (GPT-4.1 / mini / nano) · NetworkX · pandas · Matplotlib

- Replicated and extended an **ICWSM 2025 paper** (Chang et al., *LLMs generate structurally realistic social networks but overestimate political homophily*) with a full pipeline that generates, analyzes, and benchmarks LLM-synthesized social graphs.
- Designed a 4-axis experimental matrix — **4 prompting methods × 4 cultures × 4 prompt languages × 3 GPT-4.1 model tiers × 2 seeds = 96+ verified conditions** — isolating cultural, linguistic, and model effects on topology and homophily.
- Quantified inter-model divergence via pairwise edge distance (gpt-4.1 ↔ gpt-4.1-mini = 0.074; gpt-4.1 ↔ gpt-4.1-nano = 0.119) and identified **political affiliation as the dominant homophily dimension** and prop_nodes_lcc as the topology metric most sensitive to culture and language.

RUL Prediction with LSTM — Deep Learning for Predictive Maintenance — *Apr 2026*

PyTorch · TensorFlow · scikit-learn · NASA C-MAPSS · ipywidgets

- Designed and benchmarked **four LSTM architectures** (VanillaLSTM, StackedBiLSTM, LSTM-Attention, CNN-LSTM) on NASA C-MAPSS for turbofan RUL prediction.
- Implemented **Monte Carlo Dropout and Deep Ensembles** for uncertainty quantification, using mean $- 2\sigma$ for risk-adjusted maintenance decisions.
- Applied **gradient-based XAI and temporal attention heatmaps** for auditable explanations; built an ipywidgets dashboard with tiered maintenance alerts and zero-shot cross-dataset transfer.

EDUCATION

University of Arizona — M.S. in Data Science · GPA 3.77 / 4.00 — *Aug 2024 – May 2026*

SRM University, India — B.Tech in Computer Software Engineering — *2020 – 2024*